Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_20

1. Summary of Key Deviations

The analysis of Run_20 reveals a significant and complex physical fault. The primary symptom is a massive loss of combustion pressure, which occurs despite other indicators suggesting the system is attempting to compensate.

- **p_Combustion (Major):** The most significant anomaly is a major and persistent **negative offset** on the combustion pressure, which reads approximately **-39%** lower than healthy across all active states. This points to a major leak or loss of efficiency in the combustion section.
- **Flow_PRV (Moderate):** Uniquely, the `Flow_PRV` signal shows a moderate positive **offset** of **+13.8%** at the very beginning of the run (state s1), indicating the high-pressure relief valve is active even at idle.
- **Flow_Engine1-4 (Minor):** In an attempt to compensate for the low pressure, the engine flows show a slight **positive offset** of around **+4-6%**.
- **Other Signals (Minor/Nominal):** All other signals, including `Flow_Pump` and `Flow_Bypass`, are either nominal or show only minor, likely secondary, deviations.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Pressure Relief Valve (PRV)**	**Stuck Partially Open / Low Cracking Pressure**	This is a very strong hypothesis. The fact that the **PRV is flowing at idle** (state s1) is highly anomalous and indicates it's opening at a much lower pressure than it should. A leaking PRV creates a permanent loss of pressure and flow from the main system, perfectly explaining the major negative offset on **p_Combustion** (-39%). The control system's attempt to compensate by slightly increasing engine flow is a logical secondary effect.

2	**Combustion Chamber / Injector Nozzle(s)**	**Major Leak**	A major leak in the combustion chamber or at one or more nozzles would explain the -39% pressure drop. It's a very plausible physical fault. However, it doesn't as elegantly explain why the `Flow_PRV` was active at idle. A faulty PRV is a more direct explanation that covers all the evidence.
3	**p_Combustion Sensor**	**Sensor Failure (Major Negative Offset)**	It's possible the `p_Combustion` sensor has simply failed and is reading low. If this were the case, the control system would react by increasing fuel flow, which is consistent with the slightly positive `Flow_Engine` readings. However, this hypothesis doesn't explain the anomalous PRV flow at idle.
4	**Main Fuel Pump**	**Major Under-performance**	A failing pump would cause a pressure drop. However, we would expect to see a significant *negative* deviation on the `Flow_Pump` sensor, but it remains nominal. This makes a pump fault unlikely.
5	**Bypass Valve**	**Stuck Open**	A stuck-open bypass valve would cause a loss of pressure, but it would also cause a significant negative deviation on the `Flow_Bypass` sensor, which is not observed here.

3. Most Probable Hypothesis

The most probable root cause is that the **Pressure Relief Valve (PRV) is Stuck Partially Open or has a low cracking pressure.**

Reasoning: The "smoking gun" for this diagnosis is the `Flow_PRV` reading of +13.8% in state `s1` (idle). The PRV should only ever activate during a severe over-pressure event, never at idle. This indicates the valve is faulty and is leaking fuel out of the main circuit from the very beginning of the run. This single, continuous leak provides a perfect explanation for the primary physical symptom: a massive loss of system pressure (`p_Combustion` is -39%). The control system's attempt to compensate by slightly increasing engine flow is a logical, but ultimately futile, reaction to this persistent leak.